PELAYO PACHE RODRÍGUEZ

INDUSTRIAL ELECTRONICS & AUTOMATION ENGINEERING STUDENT

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Education

UNIVERSIDAD DE OVIEDO (EPI Gijón) **Gijón, Spain**

B.Sc. in Industrial Electronics and Automation Engineering Sept. 2022 – Present (Expected Jun. 2027)

- **Academic Record:** Cumulative GPA: **7.49 / 10.0** (Equivalent to UK 2:1 Honours / ∼3.2 US GPA).
- **Bachelor's Thesis (TFG):** Currently developing TFG in direct collaboration with **Universidad de León** in **Aeronautical Engineering**.
- **Relevant Coursework: Digital Electronics**, **Embedded Systems**, **Microelectronics**, **Power Electronics**, **Control Engineering**, Analog Circuit Design, Electronic Instrumentation, Industrial Automation.

LIVERPOOL JOHN MOORES UNIVERSITY (LJMU) **Liverpool, United Kingdom**

Academic Exchange Year – Faculty of Engineering and Technology *Sept. 2025 – Present*

- **Academic Classification:** Awarded **Distinction / First Class Honours** (Equivalent GPA in the Spanish academic system: **9.15 / 10.0**).
- Full academic year taught entirely in English, focusing on advanced embedded automation, sensor integration, applied control theory, and industrial systems within an international UK setting.

SANTO DOMINGO DE GUZMÁN COLLEGE (Dominicos) **Oviedo, Spain**

High School Diploma (Scientific-Technological Track) *Sept. 2020 – May 2022*

Engineering Experience & Technical Projects

RESEARCH COLLABORATION GRANT **Oviedo / Gijón, España**

Selected for Top Research Collaboration Project *Sept. 2026 – Present*

- **Innovation Selection:** Development of laboratory test equipment for satellite constellation release testing or solar panel deployment in simulated microgravity.
- **Project Defense:** Delivery of a comprehensive project report for formal defense, endorsed by the faculty advisor for submission to the Ministry of Education.

ART – ASTURIAS RURAL TECH (UNIOVI EMPRENDE) **Oviedo / Gijón, España**

Co-Fundador y Desarrollo Tecnológico (Top 12 Proyectos de Innovación) *Sept. 2026 – Presente*

- **Top Innovation Finalist Selection:** Selected as **Finalists** among the **Top 12 Innovation Projects** in the official entrepreneurship program *Uniovi Emprende*, organized in strategic partnership with **Banco Santander** and leading regional business entities; qualified for and defended the initiative before an executive evaluation tribunal on **November 5, 2026**.
- **Direct Booking Platform & Digital Architecture:** Co-designed and deployed the web platform architecture and proprietary direct booking engine (ART Portal) engineered to eliminate abusive 15%–25% commissions from multinational OTAs (Booking, Airbnb), enabling rural accommodations and commerce to retain 100% of their net margin.
- **Field Methodology («Tú a Tú») & Commercial Pilot Validation:** Spearheaded on-the-ground technical consulting and local SEO diagnostics in rural municipalities; validated the model through an operational pilot with a traditional enterprise in Pola de Laviana, achieving a **4× increase in direct customer contacts and calls** via Google Maps optimization.
- **Strategic Financing & Market Modeling:** Structured a 4-tier scalable monetization model ad-

dressing a **EUR 22M TAM** (2,500+ rural businesses in Asturias), lowering client adoption barriers to **EUR 0 upfront cost** by channeling official *Kit Digital* subsidies (EUR 3,000–EUR 12,000 grants) to foster long-term rural business sustainability.

UNIOVI ETECH-RACING (FORMULA STUDENT) **Gijón, Spain**
Low Voltage Hardware & Electronics Engineer *Sept. 2024 – May-2026*

- **PCB Design:** Designed, routed, and validated multi-layer printed circuit boards (PCBs) using **Altium Designer** (Advanced Level) for the vehicle's Low Voltage (LV) electrical architecture, ensuring high signal integrity and strict compliance with Formula Student safety regulations.
- **Industrial Communications:** Implemented and debugged **CAN bus communication** protocols (500 kbps) for real-time data exchange between the Vehicle Control Unit (ECU), telemetry unit, and Battery Management System (BMS).
- **Wiring Bench Testing:** Engineered custom modular wiring harnesses with motorsport connectors and EMI shielding. Bench testing and **tin soldering** (THT SMD).

UNIOVI INNOVATION SKILLS **Gijón, Spain**
Technical Lead & Hardware Prototyping (Winner – Most Innovative Project)Academic Year 2023 – 2024

- Spearheaded technical design of an automated industrial monitoring solution developed in partnership with municipal innovation hub *Gijón Impulsa* and elevator manufacturer *Tresa Ascensores*.
- Engineered sensor telemetry interface and predictive logic; presented functional prototype to an executive jury, winning **1st Prize for Most Innovative Project**.

Additional Work Experience & Leadership

PRIVATE TUTOR IN MATHEMATICS & PHYSICS **Oviedo, Spain**
Private Tutor *2022 – Present*

- Delivered personalized instruction in calculus, linear algebra, and classical mechanics to over 10 secondary and pre-university students, strengthening technical communication skills.

EDP (ELECTRICIDADE DE PORTUGAL) – FIDMA TRADE FAIR **Gijón, Spain**
FIDMA EDP Assistant *August 2023*

- Customer support, brand representation, and public engagement on renewable energy and transition projects at EDP's fair pavilion.

ALVAPAN **Oviedo, Spain**
Delivery Assistant (Summer Contract) *Summer 2022*

- Managed delivery logistics, route organization, and customer service under tight schedules in a demanding environment.

Technical & Social Skills

- **Systems Engineering & MBSE (HIGHLIGHTED): SysML v2** – Model-Based Systems Engineering (MBSE), with practical application both in specialized **Dassault CATIA / MagicDraw** software and in agnostic model architectures **without CATIA** (continuously advancing learning).
- **Hardware & EDA Design: Altium Designer** (Advanced), KiCAD, Multilayer PCB Design, Schematic Capture, Circuit Prototyping, **Tin Soldering** (THT SMD), Bench Equipment (Oscilloscopes, Logic Analyzers, Regulated Power Supplies).
- **Embedded Systems & Logic:** Microcontrollers (**STM32**, **ARM Cortex**, **Microchip PIC** with **MPLAB**), **VHDL (Xilinx Vivado)**, **CAN Bus**, SPI, I2C, UART, Digital Logic, Siemens PLC (**TIA Portal**).
- **Simulation & Modeling: MATLAB & Simulink**, **PLECS**, **PSIM**, AutoCAD.
- **Programming Languages: C/C++**, **Python**, **Java**, **R**.

- **Social Management Skills:** Leadership, Teamwork in demanding technical environments, Problem-solving and management skills, Technical issue analysis and resolution, Adaptability and quick learning.
- **Languages:**

  Spanish: Native / Bilingual.

  English: C1 Full Professional Fluency (Oxford ELLT C1; Cambridge B2; Cambridge IELTS C1).

  French: Basic / Very limited proficiency.

Certifications & Training

- **Cambridge IELTS (C1 Proficient):** University of Cambridge / British Council.
- **Oxford ELLT (C1 Proficient):** University of Oxford / OI Digital Institute.
- **British Airways – Engineering Job Simulation:** Aircraft mechanics, systems safety, and fleet maintenance analysis (Forage).
- **Santander Open Academy:** Artificial Intelligence & Large Language Models with Gemini.
- **Cambridge First Certificate in English (FCE - B2):** University of Cambridge.
- **Driving License:** Class B.